Chapter 7. Distributional DQN: Getting the full story

This chapter covers

- Why a full probability distribution is better than a single number
- Extending ordinary deep Q-networks to output full probability distributions over Q values
- Implementing a distributional variant of DQN to play Atari Freeway
- Understanding the ordinary Bellman equation and its distributional variant
- Prioritizing experience replay to improve training speed

We introduced Q-learning in chapter 3 as a way to determine the value of taking each possible action in a given state; the values were called action values or Q values. This allowed us to apply a policy to these action values and to choose actions associated with the highest action values. In this chapter we will extend Q-learning to not just determine a point estimate for the action values, but an entire distribution of action values for each action; this is called *distributional Q-learning*. Distributional Q-learning has been shown to result in dramatically better performance on standard benchmarks, and it also allows for more nuanced decision-making, as you will see. Distributional Q-learning algorithms, combined with some other techniques covered in this book, are currently considered a state-of-the-art advance in reinforcement learning.

Most environments we wish to apply reinforcement learning to involve some amount of randomness or unpredictability, where the rewards we observe for a given state-action pair have some variance. In ordinary Q-learning, which we might call *expected-value Q-learning*, we only learn the average of the noisy set of observed rewards. But by taking the average, we throw away valuable information about the dynamics of the environment. In some cases, the rewards observed may have a more complex pattern than just being clustered around a single value. There may be two or more clusters of different reward values for a given state-action; for example,